

```
theory Week_1_sol
imports Main
begin
```

King's College London
Department of Informatics
Dr. Mohammad Abdulaziz

Term 1

Computer-Assisted Theorem Proving: 7CCSACTP

Exercise Sheet 1

About these solutions. Every Isabelle proof method (e.g. *rule*, *simp*), Isar keyword (e.g. *have*, *also*), attribute (e.g. *OF*, *of*) and method modifier (e.g. *arbitrary:*, *simp add:*) is explained in a short comment, marked —, the first time it appears in the solutions, in sheet order from Week 1. Later uses are not explained again, so if you meet an unfamiliar construct, look for its first use in an earlier solution. Each proof is also labelled with its style; the module's preferred style is declarative (Isar), where every step states what it proves.

Important Note: Please download Isabelle from the link provided in the slides and bring a laptop to the large group tutorial. Performing the proofs in Isabelle will be an integral part of those tutorials.

Before you start, please open a new Isabelle/HOL theory file called `Week_1.thy` and add the following to its beginning.

```
theory Week_1
imports Main
begin
```

Exercise 1.1 Trying out Isabelle

In this sheet, you should try out the Isabelle theorem prover. Prove the following theorem, first on pen-and-paper, then formally in Isabelle.

```
lemma
  assumes T: T b
    and A: A a  $\wedge$  A b  $\implies$  a = b
    and TA:  $\bigwedge x. T x \implies A x$ 
```

and $Aab: A\ a$
shows $T\ a$

proof—

- *proof* — starts a structured proof without applying any method, leaving the goal unchanged.
- Style: declarative (Isar *have/show*) with apply-style steps; forward (*OF*); backward (*rule*); *subst*.

have $1: A\ b$

- *have* states an intermediate fact (here named *1*) and proves it.
- using** $TA[OF\ T]$
- *using* passes facts to the proof method that follows.
- $r[OF\ f]$ discharges the first premise of rule r with fact f (forward reasoning); here $TA[OF\ T]$ is $A\ b$.
- .
- . proves the goal directly by a fact given with *using*, instantiating its variables if needed.

have $2: A\ a \wedge A\ b$

- apply**(*rule conjI*)
- *rule* r matches the conclusion of r with the goal and replaces the goal by r 's premises (backward reasoning); here *conjI* leaves $A\ a$ and $A\ b$.
- using** $Aab\ 1$
- .

have $3: a = b$

- apply**(*rule A*)
- using** 2
- .

show *?thesis*

- *show* proves the goal of the enclosing proof; *?thesis* abbreviates that goal.
- using** T
- apply**(*subst 3*)
- *subst eq* rewrites the goal with the equation *eq*, left to right; here a becomes b .
- .

qed

- *qed* closes the proof block, once *show* has proved its goal.

end